

AMMAR ALI

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EDUCATION

New York University

Master of Science in Computer Science

Expected May 2028

Brooklyn, NY

Stony Brook University

Bachelor of Science in Computer Science, Specialization in Artificial Intelligence and Data Science | GPA: 3.91 *Stony Brook, NY*

May 2026

Courses: Software Engineering, System Fundamentals, Data Structures, Artificial Intelligence, Data Science, Scripting Languages

Honors: Presidential Scholarship, Outstanding Academic Achievement Award, Dean's List (all semesters)

SKILLS

Languages: Python, Java, C, JavaScript/TypeScript, SQL, R, C++, OCaml

ML & Data: PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Jupyter, TF-IDF, PCA, HuggingFace Transformers

AI & LLMs: RLHF, PPO, GAE, Bradley-Terry Preference Learning, Zero-shot Prompting, Claude API, Flux Dev LoRA, Replicate

Tools & Cloud: AWS (S3, App Runner, ECR), Docker, MongoDB, React, Node.js, Git, Firebase

EXPERIENCE

Automotive Ethics VIP Lab, Stony Brook University

Lead ML Engineer

Aug 2024 – Present

Stony Brook, NY

- Designed and trained a full RLHF pipeline that encodes Kantian and Utilitarian ethical frameworks into autonomous vehicle decision-making, achieving **100% accuracy on Kantian** and **90%+ on Utilitarian** ethical scenarios across 200 labeled human trajectories.
- Implemented policy, value, and reward networks (~12K parameters) from scratch in PyTorch, trained via PPO with Generalized Advantage Estimation and Bradley-Terry preference learning.
- Led a team of 7 engineers through full ML lifecycle including preference pipeline, reward shaping, hyperparameter tuning, and CARLA simulator API integration, managing delivery via Jira; preparing publication for AI ethics venues.

Suffolk County Department of Information Technology

Software & Web Development Intern

Jun 2025 – Jul 2025

Hauppauge, NY

- Evaluated and benchmarked an AI chatbot (built on Google Web Crawler infrastructure) intended for the county's public website, refining performance parameters and improving interaction quality by 50%.
- Designed and developed 3 county initiative websites; maintained 50+ web pages via the county CMS.

PROJECTS

ReminisceAI Generative AI for Dementia Care (In Development) | Claude API, Flux Dev LoRA, Replicate, AWS S3, MongoDB

- Co-founding an AI startup building a reminiscence therapy platform for dementia patients, using a multi-modal generative pipeline to transform patient memories (text prompts, photos, music) into enriched, personalized VR sessions.
- Architecting an end-to-end memory enrichment pipeline leveraging Claude API for contextual narrative generation and Flux Dev LoRA (via Replicate) for personalized image synthesis; storing and serving memory assets via AWS S3 and MongoDB.

News Headline Classifier | Python, PyTorch, Scikit-learn, TF-IDF, HuggingFace, Qwen2.5

- Implemented logistic regression from scratch with batch gradient descent, L2 regularization, and sparse matrix support, achieving **91.5% test accuracy** on 1,000 AG News headlines (World vs. Sports) using TF-IDF embeddings.
- Engineered a few-shot prompt for Qwen2.5-0.5B-Instruct (run locally via GPU on Colab) to perform few-shot classification, achieving 69.5% accuracy; implemented post-processing to map free-form LLM output to clean labels.
- Implemented K-means clustering from scratch with convergence detection and empty-cluster reinitialization; visualized TF-IDF embedding separation via PCA, confirming natural topic cluster alignment.

Draftroom Live Fantasy Baseball Platform (50+ users) | TypeScript, React, Node.js, MongoDB, AWS, Socket.io, Docker

- Built an AI-driven mock draft simulator where 11 autonomous AI teams manage budgets and roster needs, placing competitive bids against a context-aware valuation engine that dynamically re-prices players based on live league state.
- Architected a publicly documented REST API (Amethyst API) and full-stack web app on a shared TypeScript monorepo; deployed via AWS S3, App Runner, and Docker/ECR with JWT/bcrypt auth and MongoDB Atlas multi-league isolation.

Household Electricity Usage Prediction | Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn

- Merged and cleaned 8,760-record hourly weather and energy datasets across 13 features; engineered a unified daily consumption pipeline for modeling and analysis.
- Built regression and classification models to forecast usage and classify temperature days, achieving RMSE of 7.7 kW and F1 of 0.82; produced interpretable visualizations of device-level consumption patterns.